

Vortex-core size and quantum geometry in flat-band superconductors

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We investigate the real-space vortex structure of a flat-band superconductor described by the attractive Hubbard model on the Mielke checkerboard lattice. Using a momentum-space mean-field analysis, we derive a closed-form expression for the coherence length in terms of the quantum metric of the Bloch states and a particle-hole-symmetric pair density. Self-consistent Bogoliubov-de Gennes calculations on a finite disk confirm this prediction across a wide range of fillings and interaction strengths. The coherence length is minimized at half filling, diverges toward both band edges as the pair density vanishes, and depends only logarithmically on the interaction strength. These features differ qualitatively from the conventional BCS picture, which relies on a well-defined Fermi surface and a kinetic mechanism for pair transport. Instead, the vortex-core size is governed by a geometric pair mass arising from virtual interband processes encoded in the quantum metric. Our results establish the vortex core as a direct real-space manifestation of flat-band quantum geometry.

I. INTRODUCTION

Flat-band systems provide a unique setting for exploring interaction-driven quantum phenomena. Because the kinetic energy is strongly suppressed, interactions and the quantum geometry of Bloch states can dominate the low-energy physics, giving rise to unconventional forms of superconductivity, superfluidity, and transport properties [1–8]. A central question in this context is how superconducting correlations acquire a characteristic spatial scale when the underlying band is flat. In conventional Bardeen-Cooper-Schrieffer (BCS) theory, the coherence length is controlled by the Fermi velocity and therefore ultimately by band dispersion [9, 10]. In an exactly flat band, however, neither a well-defined Fermi surface nor a conventional kinetic-energy scale exists, leaving open the question of what determines the size of spatial inhomogeneities such as vortex cores.

Recent theoretical studies have identified quantum geometry as a mechanism capable of replacing band dispersion in flat-band superconductors [11–22]. Although flat bands carry no conventional kinetic energy, virtual interband processes encoded in the quantum metric generate a finite effective mass for Cooper pairs. This geometric pair mass produces a finite superfluid weight and a finite healing length, establishing a direct connection between Bloch-state geometry and macroscopic transport properties. While this connection is now well established for homogeneous systems [1–8], much less is known about its manifestation in spatially inhomogeneous situations. In particular, it remains unclear whether the coherence length predicted from bulk quantum-geometric theories survives in fully self-consistent vortex states, where order-parameter suppression, density redistribution, and finite-size effects become important [16]. Resolving this issue is essential for connecting quantum geometry to experimentally accessible real-space observables.

In this work, we address this problem by investigating a singly quantized vortex in a flat-band superconductor described by the attractive Hubbard model on the

Mielke checkerboard lattice [23]. This lattice contains an exactly flat band touching a dispersive band through a quadratic band-touching point, providing a minimal platform in which quantum-geometric effects are both analytically transparent and numerically accessible [24]. We first develop a momentum-space description and derive a closed-form expression for the coherence length in terms of the quantum metric and a particle-hole-symmetric pair density. The resulting prediction contains no adjustable parameters beyond the interaction strength, filling, and an ultraviolet cutoff fixed independently through a state-counting procedure. We then solve the Bogoliubov-de Gennes (BdG) equations self-consistently on a finite disk and extract the vortex-core size directly from the spatial profile of the order parameter.

Our calculations reveal a remarkably close agreement between the momentum-space prediction and the real-space BdG results throughout the flat-band regime. The coherence length is smallest near half filling and diverges toward both band edges as the density of mobile pairs vanishes. Moreover, its dependence on interaction strength is only logarithmic, reflecting the logarithmic enhancement of the geometric pair mass associated with the quadratic band-touching point. The remaining quantitative differences can be traced primarily to the infrared cutoff imposed by the finite system size, which limits the development of the asymptotic low-energy behavior. These findings establish the vortex-core size as a direct real-space manifestation of the quantum-geometric pair mass and demonstrate that quantum geometry can assume the role conventionally played by band dispersion in determining the spatial structure of a superconductor.

The remainder of this paper is organized as follows. In Sec. II, we introduce the momentum-space formalism, including the lattice geometry, continuum model, mean-field theory, self-consistency equations, and the resulting coherence-length expression. In Sec. III, we develop the real-space BdG formalism and obtain self-consistent vortex solutions, from which we extract the vortex-core size and compare it with the momentum-space prediction. Finally, in Sec. IV, we summarize our main find-

ings. Technical details of the Bessel-function expansion and the BdG self-consistency relations are presented in Apps. A-E.

II. MOMENTUM-SPACE FORMALISM

Our analytical framework is based on the two-band Mielke checkerboard lattice [24]. Its exactly flat lower band, together with the presence of both time-reversal and sublattice-exchange symmetries, makes it a particularly transparent platform for establishing the key concepts and deriving the closed-form results that form the foundation of the real-space BdG analysis.

A. Mielke checkerboard lattice

The Mielke checkerboard is a two-sublattice (A, B) line-graph lattice with lattice spacing a and hopping amplitude t , shown in Fig. 1(a) [23]. Its geometry enforces destructive interference between neighboring hopping paths, giving rise to an exactly flat Bloch band. In the sublattice basis $\phi_{\sigma\mathbf{k}} = (c_{A\sigma\mathbf{k}}, c_{B\sigma\mathbf{k}})^T$, where $\sigma = \{\uparrow, \downarrow\}$ labels the spin and $\mathbf{k} = (k_x, k_y)$ denotes the crystal momentum (with $\hbar = 1$), the noninteracting Hamiltonian is $\mathcal{H}_0 = \sum_{\sigma\mathbf{k}} \phi_{\sigma\mathbf{k}}^\dagger H_{\mathbf{k}} \phi_{\sigma\mathbf{k}}$. The spin-independent Bloch Hamiltonian $H_{\mathbf{k}}$ can be written as

$$H_{\mathbf{k}} = d_{\mathbf{k}}^0 \tau_0 + d_{\mathbf{k}}^x \tau_x + d_{\mathbf{k}}^z \tau_z, \quad (1)$$

where τ_0 is the identity matrix and τ_x and τ_z are Pauli matrices acting on the sublattice degree of freedom. The coefficients are $d_{\mathbf{k}}^0 = 2t[1 + \cos(k_x a) \cos(k_y a)]$, $d_{\mathbf{k}}^x = 2t[\cos(k_x a) + \cos(k_y a)]$, and $d_{\mathbf{k}}^z = 2t \sin(k_x a) \sin(k_y a)$. Since $\mathbf{d}_{\mathbf{k}} = \mathbf{d}_{-\mathbf{k}}$, the model possesses inversion symmetry, $H_{\mathbf{k}} = H_{-\mathbf{k}}$. Furthermore, the absence of a τ_y component makes $H_{\mathbf{k}}$ purely real, implying time-reversal symmetry, $H_{\mathbf{k}} = H_{-\mathbf{k}}^*$.

The Bloch-band energies are $\epsilon_{s\mathbf{k}} = d_{\mathbf{k}}^0 + s d_{\mathbf{k}}$, where $d_{\mathbf{k}} = |\mathbf{d}_{\mathbf{k}}| = 2t[1 + \cos(k_x a) \cos(k_y a)]$ and $s = \pm$. Since $d_{\mathbf{k}}^0 = d_{\mathbf{k}}$, the spectrum reduces to $\epsilon_{-, \mathbf{k}} = 0$ and $\epsilon_{+, \mathbf{k}} = 4t[1 + \cos(k_x a) \cos(k_y a)]$. Thus, the lower band is perfectly flat, whereas the upper band is dispersive, as shown in Fig. 1(b). The two bands touch quadratically at the four symmetry-related points $\mathbf{k} = (\pm\pi/a, 0)$ and $\mathbf{k} = (0, \pm\pi/a)$, where $d_{\mathbf{k}} = 0$ and therefore $\epsilon_{s\mathbf{k}} = 0$. Consequently, the flat band is not separated from the dispersive band by an energy gap, but instead forms a quadratic band touching. Despite its minimal two-band structure, the Mielke checkerboard lattice captures the essential ingredients of flat-band quantum geometry. The quadratic touching points give rise to a singular quantum metric in the long-wavelength limit, while the flat lower band suppresses conventional kinetic contributions. As a result, this model provides an ideal setting for isolating and analyzing geometric effects in pairing, superfluid transport, and collective properties of flat-band systems.

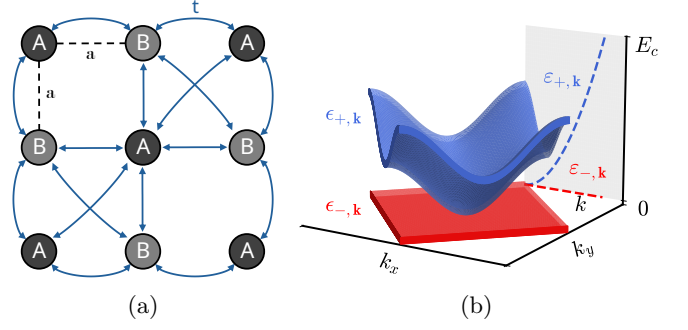


FIG. 1: (a) Schematic of the two-orbital Mielke checkerboard lattice, with sublattices labeled A and B . The lattice spacing and hopping amplitude are denoted by a and t , respectively. (b) Band structure of the lattice, showing the flat band (red) and the dispersive band (blue). The shaded region indicates the low-energy continuum sector retained in our analysis, bounded by the cutoff energy E_c and projected onto the radial momentum $k = |\mathbf{k}|$.

B. Effective continuum model

Since the flat-band physics is governed by the vicinity of the quadratic band-touching points, we formulate the remainder of this work within a low-energy continuum description obtained by expanding $H_{\mathbf{k}}$ to second order around one such point. Measuring energies relative to the touching energy, the Bloch coefficients become

$$d_{\mathbf{k}}^0 = ta^2 k^2, \quad d_{\mathbf{k}}^x = ta^2(k_x^2 - k_y^2), \quad d_{\mathbf{k}}^z = 2ta^2 k_x k_y, \quad (2)$$

with $k^2 = k_x^2 + k_y^2$. Diagonalizing the resulting Hamiltonian yields

$$\epsilon_{-, \mathbf{k}} = 0, \quad \epsilon_{+, \mathbf{k}} = 2ta^2 k^2, \quad (3)$$

so that the lower band remains perfectly flat at zero energy, while the upper band acquires an isotropic quadratic dispersion. This behavior is illustrated in the low-energy projection of Fig. 1(b).

The geometric content of the Bloch states governing interband response is encoded in the quantum metric tensor. For a two-band model, it is convenient to express it in terms of the normalized vector $\hat{\mathbf{d}}_{\mathbf{k}} = \frac{\mathbf{d}_{\mathbf{k}}}{|\mathbf{d}_{\mathbf{k}}|} = \frac{1}{k^2}(k_x^2 - k_y^2, 0, 2k_x k_y)$, from which the quantum metric follows as $g_{ij}^{\mathbf{k}} = \frac{1}{2} \partial_i \hat{\mathbf{d}}_{\mathbf{k}} \cdot \partial_j \hat{\mathbf{d}}_{\mathbf{k}}$, leading to

$$g_{ij}^{\mathbf{k}} = \frac{2\delta_{ij}}{k^2} - \frac{2k_i k_j}{k^4}, \quad (4)$$

where $\partial_i \equiv \frac{\partial}{\partial k_i}$ and δ_{ij} is the Kronecker delta. In the following sections, we consider quantities such as the coherence length and effective inverse-mass tensors, which are expressed as momentum sums of the quantum metric weighted by an isotropic function $w(k)$, $\mathcal{O}_{ij} = \sum_{\mathbf{k}} g_{ij}^{\mathbf{k}} w(k)$. The off-diagonal component contains the factor $k_x k_y w(k)/k^4$, which is odd under $k_x \rightarrow -k_x$

(or $k_y \rightarrow -k_y$) and therefore vanishes upon Brillouin-zone integration. Moreover, rotational symmetry enforces equality of the diagonal components under $k_x \leftrightarrow k_y$. Consequently, all such observables are diagonal and isotropic, $\mathcal{O}_{ij} = \mathcal{O}\delta_{ij}$, so it suffices to consider a single diagonal component (e.g., xx), allowing these quantities to be treated effectively as scalars.

C. Mean-field Hamiltonian

We now include an on-site attractive Hubbard interaction of strength $U \geq 0$ between $\uparrow$ and $\downarrow$ fermions and treat it within the BCS mean-field approximation. Owing to the sublattice-exchange symmetry of the lattice, the superconducting order parameter is uniform in space [24], i.e., $\Delta_A = \Delta_B \equiv \Delta_0$, which we take to be real without loss of generality. Introducing the four-component Nambu spinor $\Phi_{\mathbf{k}} = (\phi_{\uparrow\mathbf{k}}, \phi_{\downarrow,-\mathbf{k}}^\dagger)^T$, the mean-field Hamiltonian becomes

$$\mathcal{H}_{\text{mf}} = \mathcal{E}_0 + \sum_{\mathbf{k}} \Phi_{\mathbf{k}}^\dagger \begin{pmatrix} H_{\mathbf{k}} - \mu\tau_0 & \Delta_0\tau_0 \\ \Delta_0\tau_0 & -H_{-\mathbf{k}}^* + \mu\tau_0 \end{pmatrix} \Phi_{\mathbf{k}}, \quad (5)$$

where $\mathcal{E}_0 = N\Delta_0^2/U + 2\sum_{\mathbf{k}}(d_{\mathbf{k}}^0 - \mu)$ is a constant energy offset, with N the number of lattice sites, and μ is the chemical potential. Diagonalization yields the quasiparticle spectrum

$$E_{s\mathbf{k}} = \sqrt{\xi_{s\mathbf{k}}^2 + \Delta_0^2}, \quad \xi_{s\mathbf{k}} = \varepsilon_{s\mathbf{k}} - \mu, \quad (6)$$

which reflects the fact that uniform pairing does not mix the band index beyond the particle-hole structure. Consequently, the quasiparticle Hamiltonian remains block diagonal in the band index, which underlies the self-consistency equations derived in the following section.

D. Self-consistency equations

The pairing field and chemical potential are determined self-consistently by requiring that the order parameter and particle number obtained from the mean-field spectrum in Eq. (6) reproduce their assumed values. The resulting gap and number equations can be written compactly as [24]

$$1 = \frac{U}{2N} \sum_{s\mathbf{k}} \frac{\mathcal{X}_{s\mathbf{k}}}{E_{s\mathbf{k}}}, \quad (7)$$

$$F = 1 - \frac{1}{N} \sum_{s\mathbf{k}} \frac{\xi_{s\mathbf{k}}}{E_{s\mathbf{k}}} \mathcal{X}_{s\mathbf{k}}, \quad (8)$$

where $\mathcal{X}_{s\mathbf{k}} = \tanh[E_{s\mathbf{k}}/(2T)]$ is the thermal factor (with $k_B = 1$), and $F = \mathcal{N}/N \in [0, 2]$ denotes the filling per site, with $\mathcal{N}$ the total number of particles in the system. The BCS mean-field solution is obtained by simultaneously solving Eqs. (7) and (8) for Δ_0 and μ at fixed U ,

F , and T [25]. To evaluate the momentum sums analytically, we set $\sum_{\mathbf{k}} \rightarrow \frac{A}{(2\pi)^2} \int_{|\mathbf{k}| < k_c} d^2\mathbf{k}$, with $A = Na^2$ the total system area, which introduces an ultraviolet radial-momentum cutoff k_c .

To establish a direct correspondence with the real-space BdG calculations performed on a disk of radius R , we fix the ultraviolet cutoff by requiring that the number of states retained in the momentum-space theory equals the number of unit cells contained within the corresponding real-space disk. Specifically, the number of continuum $\mathbf{k}$ states contained within a disk of radius k_c is $N_c = \sum_{\mathbf{k}} 1 = \frac{Ak_c^2}{4\pi}$, whereas the number of unit cells within the corresponding lattice disk is $N_c = \frac{A}{2a^2}$, since each unit cell occupies an area $2a^2$, with $N = 2N_c$ and $A = \pi R^2$. Matching these two expressions yields $k_c = \frac{\sqrt{2\pi}}{a}$, which in turn determines the cutoff energy as

$$E_c = 2ta^2k_c^2 = 4\pi t, \quad (9)$$

exceeding the bandwidth $8t$ of the original lattice model. This prescription ensures that the continuum theory and the real-space BdG formulation contain exactly the same number of single-particle degrees of freedom. It therefore provides a natural bridge between the bulk momentum-space calculations and the finite-size real-space simulations discussed in Sec. III. The construction is closely analogous to the Debye prescription in lattice dynamics, where the continuum phonon spectrum is truncated such that the total number of phonon modes equals the number of microscopic lattice degrees of freedom.

The importance of this cutoff extends beyond a simple numerical convenience. Because the low-energy continuum model neglects the microscopic lattice structure at short distances, its ultraviolet sector must be constrained so that the correct state counting of the underlying Hubbard model is preserved. The present prescription achieves exactly this. In particular, it guarantees that high-energy states contribute with the correct total spectral weight, allowing quantities that are sensitive to ultraviolet physics, such as the two-body binding energy, μ and Δ_0 , to recover the correct Hubbard-model behavior [24]. As we demonstrate throughout this work, this state-counting procedure allows the continuum formalism to remain quantitatively reliable well beyond the weak-coupling regime, while providing a controlled connection between momentum-space and real-space descriptions of the same physical system.

For instance, at zero temperature ($T = 0$) and in the $F \rightarrow 0$ limit, the gap equation in Eq. (7) reduces to the two-body bound-state condition in vacuum, $\frac{1}{U} = \frac{1}{N} \sum_{s\mathbf{k}} \frac{1}{2\varepsilon_{s\mathbf{k}} - E_b}$, where $|E_b|$ with $E_b < 0$ denotes the binding energy of a spin-singlet pair at zero center-of-mass momentum [26]. Evaluating this equation within the low-energy continuum model and introducing the same radial cutoff k_c , we obtain

$$\frac{t}{U} = -\frac{E_c}{8\pi E_b} + \frac{1}{16\pi} \ln\left(1 - \frac{2E_c}{E_b}\right). \quad (10)$$

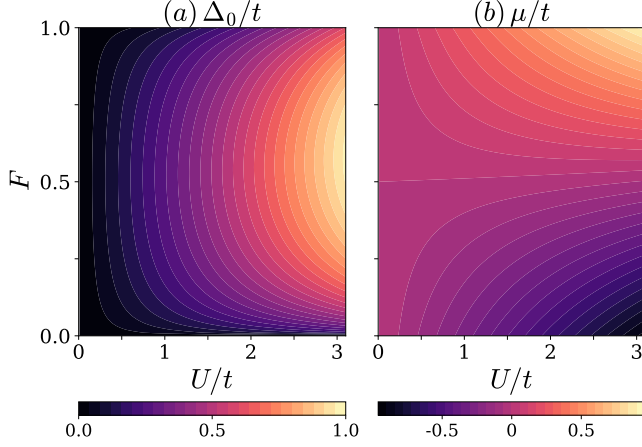


FIG. 2: Self-consistent zero-temperature solutions of (a) the pairing gap Δ_0 and (b) the chemical potential μ obtained from Eqs. (11) and (12). In the weak-coupling limit ($U/t \rightarrow 0$), both quantities exhibit exact particle-hole symmetry about $F = 0.5$, corresponding to a half-filled flat band.

This expression reproduces the expected limiting behaviors. In the weak-coupling regime $|E_b| \ll E_c$, the logarithmic contribution is subleading, and one finds $E_b \approx -U/2$, consistent with the two-particle bound-state energy of the attractive Hubbard model in this regime. In the opposite limit $|E_b| \gg E_c$, the logarithm can be expanded to leading order in $E_c/|E_b|$, yielding $E_b \approx -U$, which recovers the correct strong-coupling molecular limit of the lattice problem. The recovery of both limits is a nontrivial consequence of the cutoff prescription introduced above, which preserves the correct number of lattice degrees of freedom in the continuum description. Throughout this work, however, we focus on the flat-band regime $0 \leq F \leq 1$ with $|E_b| \ll E_c$, where the continuum approximation remains quantitatively controlled in the BdG calculations.

At $T = 0$, the thermal factor reduces to $\mathcal{X}_{s\mathbf{k}} = 1$, and the continuum momentum integrals in Eqs. (7) and (8) can be evaluated analytically. The resulting self-consistency equations take the form

$$\frac{1}{U} = \frac{1}{4E_0} + \frac{1}{16\pi t} \ln \left[\frac{E_c - \mu + \sqrt{(E_c - \mu)^2 + \Delta_0^2}}{-\mu + E_0} \right], \quad (11)$$

$$F = \frac{1}{2} + \frac{\mu}{2E_0} + \frac{E_c + E_0 - \sqrt{(E_c - \mu)^2 + \Delta_0^2}}{8\pi t}, \quad (12)$$

where $E_0 = \sqrt{\mu^2 + \Delta_0^2}$. For fixed interaction strength U and filling F , Eqs. (11) and (12) determine μ and Δ_0 , whose numerical solutions are shown in Fig. 2. As a useful reference, the noninteracting limit ($\Delta_0 = 0$) yields $F = 1/2$ at $\mu = 0$, reflecting the half-filling of the flat band, while $F = 1 + \mu/(4\pi t)$ for $\mu > 0$, corresponding to occupation of the quadratically dispersing upper

band. Further insight is obtained from the asymptotic solutions: in the weak-coupling regime $|E_b| \ll E_c$, we find $\mu = \frac{U}{2}(F - \frac{1}{2})$ and $\Delta_0 = \frac{U}{2}\sqrt{F(1-F)}$, valid for $0 \leq F \leq 1$ and exhibit particle-hole symmetry about $F = 1/2$. In the opposite regime $|E_b| \gg E_c$, where $|\mu| \gg E_c$ and $\mu < 0$, the corresponding solutions become $\mu = -\frac{U}{2}(1-F)$ and $\Delta_0 = \frac{U}{2}\sqrt{F(2-F)}$, valid for $0 \leq F \leq 2$ and exhibit particle-hole symmetry about $F = 1$, corresponding to half filling of the full two-band system. Thus, in the dilute limit $F \rightarrow 0$, both asymptotic solutions reduce to $\mu \rightarrow E_b/2$, despite being obtained from opposite coupling limits. Consequently, the many-body chemical potential is controlled entirely by the two-body bound state, indicating that the elementary low-energy degrees of freedom are pairs rather than individual fermions [10, 25]. Unlike a conventional dispersive band, where weak attraction leads to a BCS state built around a Fermi surface, the flat band lacks a kinetic-energy scale that can compete with pair formation. As a result, the dilute flat-band superfluid possesses an intrinsically molecular character and is most naturally viewed as a Bose-Einstein condensate (BEC) of bound pairs, even in the weak-coupling limit [14, 15, 26].

In Sec. III, we demonstrate that the real-space BdG calculations are in excellent agreement with the bulk self-consistent solutions of Eqs. (11) and (12). Before presenting those results, we first examine the zero-temperature coherence length within the momentum-space formalism, which provides an independent benchmark for the vortex healing length extracted from the BdG simulations.

E. Coherence length

To extract a characteristic zero-temperature length scale in the flat-band regime, we map the dilute system of Cooper pairs onto a weakly-interacting BEC [27], following Ref. [14, 21]. The corresponding healing length is [27]

$$(\xi_B^2)_{ij} = \frac{(M_B^{-1})_{ij}}{2U_B n_B}, \quad (13)$$

where M_B^{-1} is the inverse effective-pair-mass tensor (derived below), U_B is the effective pair-pair interaction, and n_B is the boson density. In the dilute limit, one may identify $U_B \simeq 2U$ and $n_B = F_B/a^2$ with $F_B \simeq F/2$ when $0 \leq F \leq 1$ and $|E_b| \ll E_c$ [14, 26]. Away from the strictly dilute regime, we employ the particle-hole symmetric interpolation $F_B = F(1-F)/2$ for the condensed pair density, which vanishes at the band edges and reproduces $F_B \simeq F/8$ near half filling, consistent with the solutions of Eqs. (11) and (12) [28]. Substituting the pair mass obtained below yields

$$\frac{\xi_B}{a} = \frac{1}{\sqrt{4F_B}} \left[\frac{1}{8\pi} + \frac{1}{4\pi} \ln \left(\frac{E_c}{U} \right) \right]^{1/2}, \quad (14)$$

which provides a simple analytical estimate for the coherence length.

Equation (14) reveals that the coherence length in a flat-band superfluid is governed primarily by filling rather than by interaction strength. Indeed, the interaction enters only through the logarithmic factor $\ln(E_c/U)$, implying that even substantial changes in U produce only modest variations in ξ_B . This behavior is in sharp contrast to conventional dispersive-band superconductors, where the coherence length is typically much more sensitive to the pairing strength. Instead, the dominant dependence arises through the bosonic density factor $1/\sqrt{F_B}$. As the flat band approaches either the empty limit ($F \rightarrow 0$) or the fully occupied limit ($F \rightarrow 1$), the density of mobile pairs vanishes, causing the healing length to diverge. Physically, the condensate then becomes increasingly dilute and loses its ability to recover from local perturbations over short distances. Conversely, the pair density is maximal at half filling, where ξ_B reaches its minimum value. This predicts that vortices and other spatial textures should be most localized near half filling and become progressively more extended as the flat band is emptied or filled. As shown in Sec. III A, these trends are borne out by fully self-consistent real-space BdG calculations, providing an important consistency check on the effective bosonic description.

The pair mass M_B is governed by the quantum geometry of the Bloch states. For instance, following Ref. [14, 26], the exact inverse effective-mass tensor of the lowest two-body bound state decomposes as $(M_{2b}^{-1})_{ij} = (M_{2b}^{-1})_{ij}^{\text{intra}} + (M_{2b}^{-1})_{ij}^{\text{inter}}$, where the intraband contribution is controlled by the band dispersion and the interband contribution by the quantum metric $g_{ij}^{\mathbf{k}}$ of Eq. (4):

$$(M_{2b}^{-1})_{ij}^{\text{intra}} = 2 \frac{\sum_{\mathbf{k}} \frac{\partial_i \varepsilon_{+, \mathbf{k}} \partial_j \varepsilon_{+, \mathbf{k}}}{(2\varepsilon_{+, \mathbf{k}} - E_b)^3}}{\sum_{s\mathbf{k}} \frac{1}{(2\varepsilon_{s\mathbf{k}} - E_b)^2}}, \quad (15)$$

$$(M_{2b}^{-1})_{ij}^{\text{inter}} = \frac{\sum_{s\mathbf{k}} \frac{g_{ij}^{\mathbf{k}}}{2\varepsilon_{s\mathbf{k}} - E_b} - \sum_{\mathbf{k}} \frac{2g_{ij}^{\mathbf{k}}}{\varepsilon_{+, \mathbf{k}} + \varepsilon_{-, \mathbf{k}} - E_b}}{\sum_{s\mathbf{k}} \frac{1}{(2\varepsilon_{s\mathbf{k}} - E_b)^2}}. \quad (16)$$

Evaluating these expressions in the weak-coupling regime relevant to the flat-band regime, where $|E_b| \ll E_c$ and $E_b \simeq -U/2$, yields $(M_{2b}^{-1})_{xx}^{\text{intra}} = \frac{Ua^2}{8\pi}$ and $(M_{2b}^{-1})_{xx}^{\text{inter}} = \frac{Ua^2}{4\pi} \ln\left(\frac{E_c}{U}\right)$. The logarithmic contribution originates from the quadratic band touching structure in two dimensions, which controls the infrared behavior of the interband processes [24, 29]. Combining the two contributions gives the isotropic inverse pair mass

$$M_{2b}^{-1} = \frac{Ua^2}{8\pi} + \frac{Ua^2}{4\pi} \ln\left(\frac{E_c}{U}\right). \quad (17)$$

Motivated by the behavior reported for the pyrochlore lattice in Ref. [14], where the Cooper-pair mass was found to coincide, to very good approximation, with the two-body bound-state mass throughout the dilute flat-band regime, we identify the pair mass M_B entering the coherence length with M_{2b} .

Equation (17) illustrates a central feature of flat-band superconductivity: although the flat band itself carries no conventional kinetic energy, bound pairs nevertheless acquire a finite mobility through virtual interband processes encoded in the quantum metric. Indeed, for $|E_b| \ll E_c$, the logarithmically enhanced interband contribution exceeds the intraband term, implying that the pair mass is controlled predominantly by the geometry of the Bloch states rather than by the dispersion of the upper band. This provides a microscopic realization of the broader principle that quantum geometry can replace band dispersion as the source of transport in flat-band systems.

The momentum-space estimate of Eq. (14) therefore provides a parameter-free prediction for the coherence length. In the next section, we test this prediction directly by solving the BdG equations in real space and extracting the vortex-core size from the self-consistent spatial profile of the order parameter.

III. REAL-SPACE FORMALISM

The momentum-space formalism of Sec. II provides an efficient description of homogeneous bulk properties but is not suitable for spatially varying order parameters. To resolve the real-space structure of the superfluid and independently benchmark the bulk predictions, we therefore employ a self-consistent BdG approach [30–32]. A particularly important application is a singly quantized vortex, $\Delta(\mathbf{r}) = \Delta(r)e^{-i\theta}$, whose phase winding and suppressed core explicitly break translational invariance on the scale of the coherence length ξ_B . Consequently, momentum is no longer a good quantum number, and a real-space formulation is required. Throughout this section, we solve the BdG equations self-consistently on a finite disk in the presence of spatially inhomogeneous pairing fields.

A. BdG equations

We retain the on-site attractive Hubbard interaction $U > 0$ and the uniform-sublattice pairing $\Delta_A = \Delta_B \equiv \Delta_0$ established in Sec. II C, but now allow the order parameter to vary in space, $\Delta_0 \rightarrow \Delta(\mathbf{r})$. The resulting mean-field Hamiltonian is

$$\mathcal{H}_{\text{mf}} = \int d^2\mathbf{r} \sum_{\sigma SS'} \psi_{S\sigma}^\dagger(\mathbf{r}) [H_{SS'}(\mathbf{r}) - \mu\delta_{SS'}] \psi_{S'\sigma}(\mathbf{r}) + \int d^2\mathbf{r} \sum_S [\Delta(\mathbf{r})\psi_{S\uparrow}^\dagger(\mathbf{r})\psi_{S\downarrow}^\dagger(\mathbf{r}) + \text{H.c.}], \quad (18)$$

where $S, S' \in \{A, B\}$ label the sublattices. The single-particle operator $H_{SS'}(\mathbf{r})$ follows from the continuum Bloch Hamiltonian through the substitution $\mathbf{k} \rightarrow -i\nabla$.

Using Eq. (2), we obtain

$$H(\mathbf{r}) = -ta^2 \begin{pmatrix} \partial_x^2 + \partial_y^2 + 2\partial_x\partial_y & \partial_x^2 - \partial_y^2 \\ \partial_x^2 - \partial_y^2 & \partial_x^2 + \partial_y^2 - 2\partial_x\partial_y \end{pmatrix} \quad (19)$$

in the sublattice basis. This operator constitutes the real-space representation of the same low-energy continuum theory used in Sec. II, preserving its two-sublattice structure and quantum-geometric content while allowing for spatially inhomogeneous pairing fields.

We diagonalize $\mathcal{H}_{\text{mf}}$ through the Bogoliubov transfor-

mation [30]

$$\psi_{S\uparrow}(\mathbf{r}) = \sum_n [u_n^S(\mathbf{r})\gamma_{n\uparrow} - v_n^{S*}(\mathbf{r})\gamma_{n\downarrow}^\dagger], \quad (20)$$

$$\psi_{S\downarrow}(\mathbf{r}) = \sum_n [u_n^S(\mathbf{r})\gamma_{n\downarrow} + v_n^{S*}(\mathbf{r})\gamma_{n\uparrow}^\dagger], \quad (21)$$

where $\gamma_{n\sigma}$ annihilates a quasiparticle in eigenstate n with spin σ , and $u_n^S(\mathbf{r})$ and $v_n^S(\mathbf{r})$ are the corresponding particle and hole amplitudes on sublattice S . Requiring the transformation to diagonalize the mean-field Hamiltonian, $[\mathcal{H}_{\text{mf}}, \gamma_{n\sigma}] = -E_{n\sigma}\gamma_{n\sigma}$ leads to the BdG eigenvalue problem

$$\begin{bmatrix} H_{AA}(\mathbf{r}) - \mu & H_{AB}(\mathbf{r}) & \Delta(\mathbf{r}) & 0 \\ H_{BA}(\mathbf{r}) & H_{BB}(\mathbf{r}) - \mu & 0 & \Delta(\mathbf{r}) \\ \Delta^*(\mathbf{r}) & 0 & -H_{AA}^*(\mathbf{r}) + \mu & -H_{AB}^*(\mathbf{r}) \\ 0 & \Delta^*(\mathbf{r}) & -H_{BA}^*(\mathbf{r}) & -H_{BB}^*(\mathbf{r}) + \mu \end{bmatrix} \begin{bmatrix} u_n^A(\mathbf{r}) \\ u_n^B(\mathbf{r}) \\ v_n^A(\mathbf{r}) \\ v_n^B(\mathbf{r}) \end{bmatrix} = E_n \begin{bmatrix} u_n^A(\mathbf{r}) \\ u_n^B(\mathbf{r}) \\ v_n^A(\mathbf{r}) \\ v_n^B(\mathbf{r}) \end{bmatrix}, \quad (22)$$

which acts in the combined sublattice and particle-hole space. The BdG Hamiltonian possesses an intrinsic particle-hole symmetry: if (u_n^S, v_n^S) is a solution at energy $E_{n\uparrow}$, then $(v_n^{S*}, -u_n^{S*})$ is a solution at energy $-E_{n\downarrow}$. We exploit this symmetry by solving only the spin- $\uparrow$ sector, henceforth denoting $E_{n\uparrow} \equiv E_n$ and allowing E_n to take both signs. Positive-energy solutions ($E_n > 0$) describe spin- $\uparrow$ quasiparticles directly, while each negative-energy solution ($E_n < 0$) is identified, via the particle-hole symmetry, with a positive-energy spin- $\downarrow$ quasiparticle at energy $E_{n\downarrow} = -E_n > 0$. The full quasiparticle spectrum for both spin species is therefore contained in the single set of eigenvalues $\{E_n\}$, with no separate diagonalization of the spin- $\downarrow$ sector required.

The pairing field must be determined self-consistently from the quasiparticle eigenfunctions. Substituting the Bogoliubov transformation into $\Delta(\mathbf{r}) = U\langle\psi_{S\downarrow}(\mathbf{r})\psi_{S\uparrow}(\mathbf{r})\rangle$ and evaluating the thermal average with the Fermi distribution $f(E) = 1/(e^{E/T} + 1)$ yields

$$\Delta(\mathbf{r}) = -U \sum_n u_n^S(\mathbf{r})v_n^{S*}(\mathbf{r})f(E_n), \quad (23)$$

which is independent of the choice of sublattice owing to the uniform on-site pairing interaction. Similarly, $\rho_\sigma(\mathbf{r}) = \sum_S \langle\psi_{S\sigma}^\dagger(\mathbf{r})\psi_{S\sigma}(\mathbf{r})\rangle$ yields the local spin densities as

$$\rho_\uparrow(\mathbf{r}) = \sum_{nS} |u_n^S(\mathbf{r})|^2 f(E_n), \quad (24)$$

$$\rho_\downarrow(\mathbf{r}) = \sum_{nS} |v_n^S(\mathbf{r})|^2 [1 - f(E_n)], \quad (25)$$

from which the total density follows as $\rho(\mathbf{r}) = \rho_\uparrow(\mathbf{r}) + \rho_\downarrow(\mathbf{r})$. The chemical potential is adjusted such that the global filling $F = \frac{1}{N} \int d^2\mathbf{r} \rho(\mathbf{r})$ matches the target value,

ensuring consistency with Eq. (8). The coupled BdG and self-consistency equations, Eqs. (22)-(25), are then solved iteratively until both the order parameter and particle number have converged.

B. Vortex profiles

We solve the BdG eigenvalue problem of Eq. (22) on a disk of radius $R = 45a$ by expanding the four-component spinor in a Bessel-function basis adapted to the disk geometry [31, 32], as detailed in Appendix A. The basis is truncated at $M = 60$ angular-momentum channels and $J = 50$ radial modes per channel, retaining only those modes whose single-particle energy lies below the cutoff $E_c = 4\pi t$ of Sec. IID, as expressed by Eq. (A7). This is the same state-counting prescription that fixes the momentum-space cutoff, so the real-space and momentum-space calculations describe the same low-energy sector and can be compared without further adjustment. The construction and diagonalization of the BdG matrix are GPU-accelerated via the cuBLAS and cuSOLVER libraries. The spectrum and pairing field are made consistent with each other iteratively using the algorithm described in Fig. 3.

We now present the self-consistent spatial profiles of $\Delta(\mathbf{r})$ and $\rho(\mathbf{r})$. We first examine the interaction dependence at fixed filling, followed by the filling dependence at fixed interaction strength. Figure 4(a) shows $\Delta(r)$ for $U/t = 0.25-1$ at fixed $F = 0.49$. In all cases, the order parameter vanishes at the vortex center due to the imposed 2π phase winding and recovers to a uniform bulk value within a few lattice spacings. The asymptotic values coincide very well with the momentum-space solutions Δ_0 obtained from Eqs. (11) and (12), as indicated by the

```

1: Initialize  $\Delta^{(0)}(r)$  from the weak-coupling momentum-
   space solution of Sec. IID.
2: for  $i = 1, 2, \dots$  until convergence do
3:   Build  $\mathcal{H}_{\text{mf}}[\Delta^{(i-1)}]$  in the Bessel basis.
4:   Diagonalize to obtain  $\{E_n, [u_n^S(r), v_n^S(r)]\}$ .
5:   Update  $\Delta^{(i)}(r)$  via Eq. (23).
6:   if  $\max_r |\Delta^{(i)} - \Delta^{(i-1)}| / \max_r |\Delta^{(i-1)}| < \epsilon$  then
7:     break
8:   end if
9: end for
10: return converged  $\Delta(r)$  and  $\rho(r)$ .

```

FIG. 3: Algorithm for the self-consistent BdG solver.

dotted horizontal black lines, providing a direct consistency check between the real-space BdG calculations and the bulk momentum-space theory. The corresponding density profiles are shown in Fig. 4(b). While the order parameter is completely suppressed at the vortex center, the particle density remains finite and approaches the bulk value $\rho_0 = F/a^2$ away from the core. This behavior reflects the presence of localized bound states that partially fill the vortex core [33]. The density depletion nevertheless becomes progressively stronger with increasing U [32], as highlighted in the inset. Thus, increasing the interaction strength primarily deepens the core depletion without qualitatively altering the vortex structure, which remains much more pronounced in $\Delta(r)$ than in $\rho(r)$.

The filling dependence at fixed $U/t = 0.5$ is shown in Fig. 5. Figure 5(a) shows the normalized profiles $\Delta(r)/\Delta_0$, which remove the trivial variation of the bulk order parameter Δ_0 and isolate the evolution of the vortex-core size. The core is most localized near half filling and expands as F approaches either band edge. Moreover, the profiles for F and $1 - F$ nearly collapse, reflecting the approximate particle-hole symmetry of Eqs. (11) and (12) about $F = 1/2$ in the weak-coupling regime. The density profiles in Fig. 5(b) reveal the same symmetry more directly [34, 35]. For $F < 1/2$, the vortex core is depleted, with $\rho(r)$ falling below its bulk value at the origin and recovering outward. For $F > 1/2$, the behavior is reversed and the core exhibits an excess density relative to the bulk. These two cases are related by the particle-hole transformation $\rho(\mathbf{r}) \rightarrow a^{-2} - \rho(\mathbf{r})$ under $F \rightarrow 1 - F$, implying that the profile at filling $1 - F$ is the reflection of that at filling F about the line $\rho a^2 = 1/2$. The near-perfect correspondence between particle-hole conjugate fillings shows that this symmetry remains quite robust even at $U = 0.5t$.

Taken together, Figs. 4 and 5 identify filling, rather than interaction strength, as the primary control parameter for the vortex structure. The relevant density scale is not F itself but the particle-hole-symmetric combination $F_B = F(1 - F)/2$, which vanishes at the band edges and is maximal at half filling. This is precisely the combination that enters the momentum-space coherence length in Eq. (14), providing a direct connection between the real-space vortex-core size and the quantum-geometric

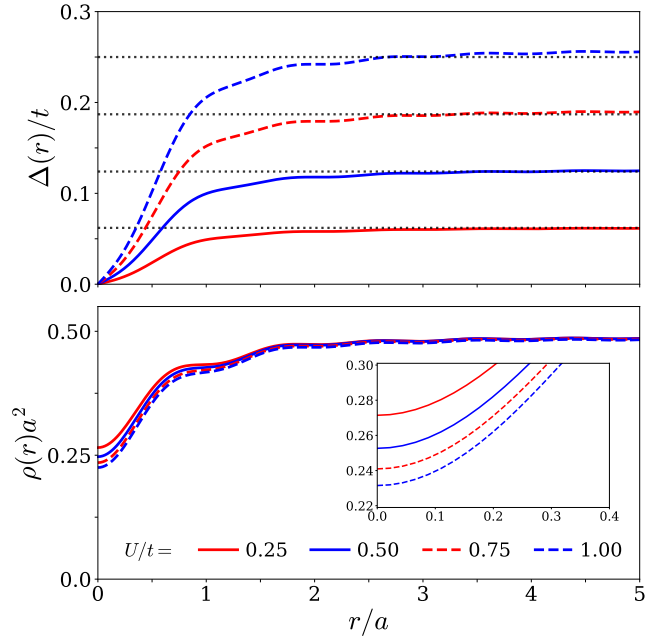


FIG. 4: (a) Radial profile of the superconducting order parameter $\Delta(r)$ for several interaction strengths U . The corresponding bulk momentum-space values are indicated by dotted horizontal black lines for comparison. (b) Local particle density $\rho(r)$ for the same parameters. The inset shows a zoom near the vortex center ($r = 0$), highlighting the finite core density arising from localized bound states. All results are obtained at fixed bulk filling $F = 0.49$.

description developed in Sec. II E.

C. Vortex-core size

To compare quantitatively with the momentum-space prediction for ξ_B , we extract a coherence length scale ξ_v from each converged order-parameter profile. In the dilute limit, the recovery of $\Delta(r)$ is well described by the bosonic healing form $\Delta(r) = \Delta_0 \tanh[r/(\sqrt{2}\xi_v)]$ [36]. See also Ref. [37]. Away from this limit, however, the vortex profile evolves with filling, making a fit to a fixed functional form unreliable. We therefore adopt a fit-free definition consistent with the same convention: ξ_v is the radius at which

$$\Delta(\xi_v) = \Delta_0 \tanh\left(\frac{1}{\sqrt{2}}\right) \approx 0.61\Delta_0, \quad (26)$$

i.e., the value attained at $r = \xi_v$ by the bosonic healing profile. This prescription recovers the conventional healing length whenever the tanh form applies and remains well defined for arbitrary profile shapes, so the comparison between the BdG results and the momentum-space prediction of Eq. (14) in Fig. 6 introduces no additional fitting parameters.

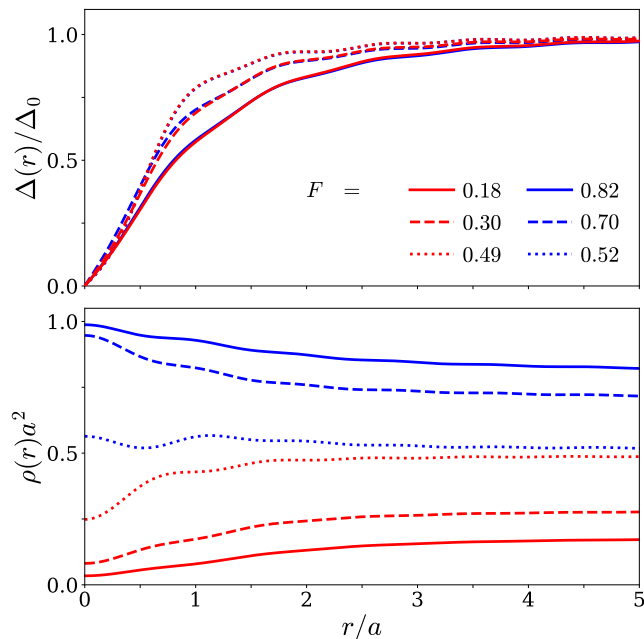


FIG. 5: (a) Normalized superconducting order parameter $\Delta(r)/\Delta_0$ for different global fillings F , illustrating the evolution of the vortex-core size with particle density. The fillings are chosen with respect to the particle-hole symmetric point. (b) Corresponding local particle density $\rho(r)a^2$ across the same filling values, computed at fixed interaction strength $U = 0.5t$.

Figure 6(a) shows ξ_v/a as a function of filling at fixed $U = 0.5t$. Two features predicted by the $1/\sqrt{F_B}$ scaling of Eq. (14) are clearly visible: a symmetric minimum at half-filling, where ξ_v falls below one lattice constant, and a divergence toward both band edges as the effective pair density vanishes. The BdG results follow the momentum-space prediction throughout the flat band, including the particle-hole symmetry about $F = 1/2$ already evident in the collapsed profiles of Fig. 5(a). This agreement is parameter-free: the analytical curve is fixed entirely by U , F , and the cutoff $E_c = 4\pi t$, with the filling dependence entering only through the particle-hole-symmetric combination $F_B = F(1 - F)/2$.

Figure 6(b) shows ξ_v/a as a function of U/t at fixed $F = 0.49$. Both descriptions yield a weak dependence, as anticipated from the logarithmic factor in Eq. (14): the BdG coherence length remains close to $0.7a$ over more than a decade in coupling strength, while the momentum-space estimate decreases only slowly with increasing U . See also Ref. [16]. Quantitative agreement is less precise here; the analytical curve overestimates the core size at weak coupling and intersects the BdG data near $U \sim t$. The residual discrepancy is attributable to finite-size effects in the BdG calculation. The logarithmic factor in Eq. (14) originates from the quadratic band touching, whose infrared contribution is cut off only at vanishing energy in the thermodynamic limit. On the finite disk,

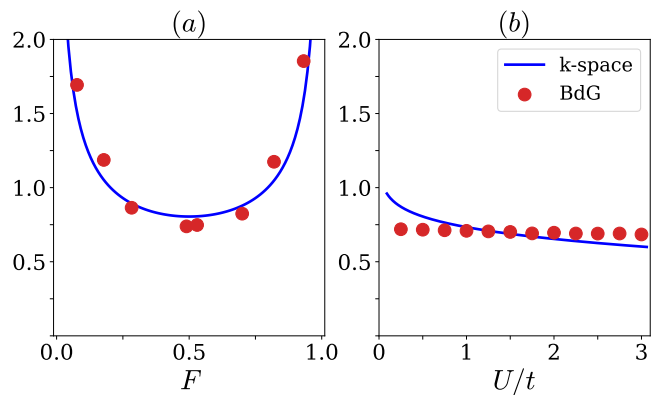


FIG. 6: Coherence length from real-space BdG calculations (ξ_v , markers) compared with the momentum-space estimate of Eq. (14) (ξ_B , solid line). (a) Filling dependence at fixed $U = 0.5t$: ξ_v/a shows a pronounced minimum at half-filling ($F = 0.5$) and diverges toward both band edges ($F \rightarrow 0$ or 1), consistent with the $F_B^{-1/2}$ scaling of Eq. (14). (b) Interaction dependence at fixed $F = 0.49$: ξ_v/a varies only weakly with U/t , reflecting the logarithmic factor in Eq. (14).

however, the single-particle spectrum is discrete and the lowest level lies a finite distance from the band-touching point, introducing an effective infrared cutoff. This suppresses the asymptotic logarithmic growth of the pair mass, and hence of ξ_v , most strongly at weak coupling, where the continuum prediction is most sensitive to low-energy states. The filling dependence, by contrast, is governed primarily by the particle-hole-symmetric factor F_B and is therefore much less affected by the discrete spectrum.

The two panels of Fig. 6 together identify the physical origin of the vortex-core size in this flat-band superconductor. The standard BCS estimate $\xi_v \sim v_F/\Delta_0$ [9, 10] breaks down because an isolated flat band lacks a well-defined Fermi surface and therefore the conventional intraband kinetic processes that govern Cooper pair transport. Instead, the finite coherence length is set by the pair mass, whose dominant contribution arises from virtual interband processes encoded in the quantum metric of the Bloch states, Eq. (4). The vortex core therefore provides a direct real-space signature of quantum geometry: its size is governed by the geometric pair mass, its weak logarithmic sensitivity to U reflects the logarithmic growth of that mass, and its strong filling dependence tracks the particle-hole-symmetric pair density F_B .

IV. CONCLUSION

We have studied the real-space structure of a singly quantized vortex in a flat-band superconductor described by the attractive Hubbard model on the Mielke checkerboard lattice. Using a self-consistent BdG formulation on

a finite disk, we computed the spatial profiles of the order parameter and particle density across a wide range of fillings and interaction strengths, and extracted a coherence length ξ_v from each converged profile using a fit-free prescription based on the bosonic healing form. We compared ξ_v against a closed-form momentum-space estimate ξ_B derived by mapping the dilute flat-band superfluid onto a weakly-interacting BEC, with the pair mass determined by the quantum metric of the Bloch states and no adjustable parameters. The two estimates agree well throughout the flat-band regime. Both independently exhibit a pronounced minimum at half-filling, diverge toward the band edges as the particle-hole-symmetric pair density $F_B = F(1 - F)/2$ vanishes, and depend on interaction strength only logarithmically. The filling dependence is well captured by the $1/\sqrt{F_B}$ scaling of the momentum-space prediction; the residual discrepancy at weak coupling is consistent with finite-size effects in the BdG calculation, which introduce an effective infrared cutoff absent in the bulk theory.

These results identify quantum geometry as the physical origin of the vortex-core size in this system. Because an isolated flat band lacks a well-defined Fermi surface, the standard BCS estimate $\xi_v \sim v_F/\Delta_0$ does not apply. Instead, ξ_v is set by the geometric pair mass, whose dominant contribution arises from virtual interband transitions encoded in the quantum metric. The vortex core therefore constitutes a direct real-space probe of flat-band quantum geometry: its size tracks the pair density through F_B , while its weak sensitivity to the pairing strength reflects the logarithmic growth of the geometric pair mass. Several directions are worth pursuing. It would be interesting to test whether this geometric mechanism persists in lattices with richer band structures or in systems where the flat band is separated from dispersive bands by a finite gap. In addition, a systematic finite-size scaling study of the weak-coupling discrepancy between ξ_v and ξ_B could help clarify the interplay between the infrared structure associated with the quadratic band touching and the discrete spectrum of a finite system.

ACKNOWLEDGMENTS

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Appendix A: Bessel-function expansion

In this appendix we derive the Bessel-function basis representation of the BdG Hamiltonian on a finite disk [31, 32]. Since the vortex solutions considered in the main text possess cylindrical symmetry, it is natural to work in polar coordinates and expand the quasiparticle wavefunctions in a basis of Bessel functions with definite angular momentum. We therefore begin by rewriting the

continuum single-particle Hamiltonian of Eq. (19) in polar form.

Introducing the radial-angular differential operators

$$\hat{P} = \partial_r^2 - \frac{1}{r}\partial_r - \frac{1}{r^2}\partial_\theta^2, \quad \hat{Q} = \frac{2}{r}\partial_r\partial_\theta - \frac{2}{r^2}\partial_\theta, \quad (\text{A1})$$

the Cartesian derivatives can be expressed as $\nabla^2 = \partial_r^2 + \frac{1}{r}\partial_r + \frac{1}{r^2}\partial_\theta^2$ for the Laplacian, $\partial_x^2 - \partial_y^2 = \cos(2\theta)\hat{P} - \sin(2\theta)\hat{Q}$ in the off-diagonal terms, and $2\partial_x\partial_y = \sin(2\theta)\hat{P} + \cos(2\theta)\hat{Q}$ in the diagonal terms. Substituting these identities into Eq. (19) yields

$$H(\mathbf{r}) = -ta^2 \begin{pmatrix} \nabla^2 + C_\theta\hat{Q} + S_\theta\hat{P} & C_\theta\hat{P} - S_\theta\hat{Q} \\ C_\theta\hat{P} - S_\theta\hat{Q} & \nabla^2 - C_\theta\hat{Q} - S_\theta\hat{P} \end{pmatrix}, \quad (\text{A2})$$

where $C_\theta = \cos(2\theta)$ and $S_\theta = \sin(2\theta)$. In this form the angular dependence is fully explicit: the Laplacian is rotationally invariant and diagonal in angular momentum, while the $\cos(2\theta)$ and $\sin(2\theta)$ factors carry angular momentum ± 2 , so the single-particle Hamiltonian couples only channels whose angular-momentum quantum numbers differ by two units. This selection rule is a direct manifestation of the quadratic band-touching structure of the continuum model and leads to a sparse block structure in the BdG matrix. As shown below, it allows the BdG differential equations to be projected efficiently onto a Bessel-function basis, reducing the problem to a finite-dimensional matrix eigenvalue equation.

We solve the BdG problem on a disk of radius R with Dirichlet boundary conditions, requiring all quasiparticle amplitudes to vanish at the edge of the system. A convenient orthonormal basis is provided by the disk eigenfunctions of the Laplacian, $\langle \mathbf{r} | jm \rangle = \frac{e^{im\theta}}{\sqrt{2\pi}} \Phi_{jm}(r)$, labeled by angular momentum m and radial index j , where

$$\Phi_{jm}(r) = \frac{\sqrt{2}}{RJ_{m+1}(\beta_{jm})} J_m\left(\beta_{jm} \frac{r}{R}\right), \quad (\text{A3})$$

Here, J_m is the Bessel function of the first kind and β_{jm} denotes its j th positive zero. The angular and radial factors satisfy the orthonormality relations

$$\int_0^{2\pi} \frac{d\theta}{2\pi} e^{i(m'-m)\theta} = \delta_{mm'}, \quad (\text{A4})$$

$$\int_0^R r dr \Phi_{jm}(r) \Phi_{j'm}(r) = \delta_{jj'}, \quad (\text{A5})$$

together with the boundary condition $\Phi_{jm}(R) = 0$ and the Laplacian eigenvalue equation

$$\nabla^2 [\Phi_{jm}(r) e^{im\theta}] = -\left(\frac{\beta_{jm}}{R}\right)^2 \Phi_{jm}(r) e^{im\theta}. \quad (\text{A6})$$

The basis therefore diagonalizes the rotationally invariant part of the single-particle Hamiltonian and is particularly well suited to the cylindrical geometry of the vortex problem. Moreover, the nontrivial angular dependence of

Eq. (A2) couples only angular-momentum channels differing by two units, so the resulting matrix representation remains sparse.

In practice the basis must be truncated. We retain angular-momentum channels $m \in [-M, M]$ and radial quantum numbers $j \in [1, J]$, keeping only those basis states whose single-particle energies lie below the continuum cutoff $E_c = 4\pi t$ derived in Sec. IID. Using the Laplacian eigenvalue β_{jm}^2/R^2 , this requirement becomes

$$\beta_{jm}^2 \leq 4\pi \left(\frac{R}{a}\right)^2. \quad (\text{A7})$$

This truncation ensures that the real-space BdG calculation contains the same number of low-energy degrees of freedom as the continuum momentum-space theory. In this study we use $M = 60$, $J = 50$, and $R/a = 45$. For these parameters the highest retained radial mode in the $m = 0$ sector satisfies $\beta_{j0}^2 \approx 4\pi(R/a)^2$, indicating that the basis nearly saturates the physical cutoff.

Appendix B: BdG Hamiltonian

Using the Bessel basis introduced in Appendix A, we now construct the explicit matrix representation of the BdG Hamiltonian (22). The particle and hole amplitudes, organized as sublattice spinors $\mathbf{u}_n = (u_n^A, u_n^B)^T$ and $\mathbf{v}_n = (v_n^A, v_n^B)^T$, are expanded in the basis of Eq. (A3) as

$$\begin{aligned} \mathbf{u}_n(\mathbf{r}) &= \sum_{m=-M}^M \sum_{j=1}^J \mathbf{u}_{njm} \frac{e^{im\theta}}{\sqrt{2\pi}} \Phi_{jm}(r), \\ \mathbf{v}_n(\mathbf{r}) &= \sum_{m=-M}^M \sum_{j=1}^J \mathbf{v}_{njm} \frac{e^{im\theta}}{\sqrt{2\pi}} \Phi_{jm}(r), \end{aligned} \quad (\text{B1})$$

with two-component sublattice coefficients $\mathbf{u}_{njm} = (u_{njm}^A, u_{njm}^B)^T$ and $\mathbf{v}_{njm} = (v_{njm}^A, v_{njm}^B)^T$.

At fixed angular momentum m , the radial and sublattice components are assembled into $2J$ -dimensional vectors,

$$\mathbf{u}_{nm} = \begin{pmatrix} \mathbf{u}_{n1m} \\ \mathbf{u}_{n2m} \\ \vdots \\ \mathbf{u}_{nJm} \end{pmatrix}, \quad \mathbf{v}_{nm} = \begin{pmatrix} \mathbf{v}_{n1m} \\ \mathbf{v}_{n2m} \\ \vdots \\ \mathbf{v}_{nJm} \end{pmatrix}. \quad (\text{B2})$$

Collecting all m -blocks yields the full particle and hole vectors

$$\mathbf{U}_n = \begin{pmatrix} \mathbf{u}_{n,-M} \\ \vdots \\ \mathbf{u}_{n,M} \end{pmatrix}, \quad \mathbf{V}_n = \begin{pmatrix} \mathbf{v}_{n,-M} \\ \vdots \\ \mathbf{v}_{n,M} \end{pmatrix}, \quad (\text{B3})$$

together with the full BdG spinor $\Psi_n = (\mathbf{U}_n, \mathbf{V}_n)^T$. With this ordering, the BdG eigenvalue problem $\mathbb{H}_{\text{BdG}} \Psi_n =$

$E_n \Psi_n$ takes the compact form

$$\mathbb{H}_{\text{BdG}} = \begin{pmatrix} \mathbf{H} - \mu \mathbf{I} & \Delta \\ \Delta^\dagger & -\mathbf{H}^* + \mu \mathbf{I} \end{pmatrix}, \quad (\text{B4})$$

where $\mathbf{I}$ is the identity in the combined radial, angular-momentum, and sublattice space. The projected single-particle and pairing matrices are indexed by m -blocks,

$$\mathbf{H} = \begin{pmatrix} \mathbf{h}_{-M,-M} & \mathbf{h}_{-M,-M+1} & \cdots & \mathbf{h}_{-M,M} \\ \mathbf{h}_{-M+1,-M} & \mathbf{h}_{-M+1,-M+1} & \cdots & \mathbf{h}_{-M+1,M} \\ \vdots & \vdots & \ddots & \vdots \\ \mathbf{h}_{M,-M} & \mathbf{h}_{M,-M+1} & \cdots & \mathbf{h}_{M,M} \end{pmatrix}, \quad (\text{B5})$$

$$\Delta = \begin{pmatrix} \Delta_{-M,-M} & \Delta_{-M,-M+1} & \cdots & \Delta_{-M,M} \\ \Delta_{-M+1,-M} & \Delta_{-M+1,-M+1} & \cdots & \Delta_{-M+1,M} \\ \vdots & \vdots & \ddots & \vdots \\ \Delta_{M,-M} & \Delta_{M,-M+1} & \cdots & \Delta_{M,M} \end{pmatrix}. \quad (\text{B6})$$

Each block $\mathbf{h}_{mm'}$ and $\Delta_{mm'}$ is a $2J \times 2J$ matrix carrying both radial (jj') and sublattice (SS') indices, so that the total dimension of $\mathbb{H}_{\text{BdG}}$ is $4(2M+1)J$, where the factor of 4 reflects the product of the particle-hole and sublattice degrees of freedom.

The block structure of Eqs. (B5)-(B6) is sharply constrained by the angular selection rule of Eq. (A4). The single-particle Hamiltonian Eq. (A2) contains only the harmonics $\{1, e^{\pm 2i\theta}\}$, restricting its matrix elements to $m' - m \in \{0, \pm 2\}$, while the singly quantized vortex pairing $\Delta(\mathbf{r}) = \Delta(r)e^{-i\theta}$ carries the single harmonic $e^{-i\theta}$, restricting its matrix elements to $m' = m + 1$. Hence

$$\mathbf{h}_{mm'} = \mathbf{0} \quad \text{unless} \quad m' = m \text{ or } m \pm 2, \quad (\text{B7})$$

$$\Delta_{mm'} = \mathbf{0} \quad \text{unless} \quad m' = m + 1, \quad (\text{B8})$$

so that $\mathbf{H}$ is block-tridiagonal in m (with even and odd m decoupled) and Δ has a single upper diagonal. These sparsity constraints reduce the number of nonzero blocks per row from $\mathcal{O}(M)$ to $\mathcal{O}(1)$, making the matrix construction and the matrix-vector products required during iterative diagonalization significantly cheaper than a dense treatment would be.

Appendix C: Single-particle Hamiltonian

We now evaluate the nonzero blocks of $\mathbf{H}$ identified by the selection rule (B7), working with the polar-form Hamiltonian of Eq. (A2) and the Bessel basis of Eq. (A3).

a. Diagonal block ($m' = m$). Only the ∇^2 term in Eq. (A2) survives the angular integration, and by Eq. (A6) it is diagonal in both the radial and sublattice indices,

$$\mathbf{h}_{mm}^{jj'} = ta^2 \frac{\beta_{jm}^2}{R^2} \tau_0 \delta_{jj'}, \quad (\text{C1})$$

where τ_0 is the 2×2 identity matrix in sublattice space.

b. Off-diagonal block ($m' = m+2$). The off-diagonal blocks of $\mathbf{H}$ are determined by the matrix elements of the two Cartesian operators $2\partial_x\partial_y$ and $\partial_x^2 - \partial_y^2$ that appear alongside ∇^2 in Eq. (A2). Using the operators $\hat{P}$ and $\hat{Q}$ defined in Eq. (A1), we introduce

$$A_{m,m+2}^{jj'} \equiv \langle jm | C_\theta \hat{Q} + S_\theta \hat{P} | j', m+2 \rangle, \quad (\text{C2})$$

$$B_{m,m+2}^{jj'} \equiv \langle jm | C_\theta \hat{P} - S_\theta \hat{Q} | j', m+2 \rangle. \quad (\text{C3})$$

The angular integrals needed to evaluate these matrix elements are

$$\int_0^{2\pi} \frac{d\theta}{2\pi} e^{2i\theta} \sin(2\theta) = \frac{i}{2}, \quad \int_0^{2\pi} \frac{d\theta}{2\pi} e^{2i\theta} \cos(2\theta) = \frac{1}{2}, \quad (\text{C4})$$

both of which follow from Eq. (A4). Acting on $\Phi_{j',m+2}(r)e^{i(m+2)\theta}$, the operators of Eq. (A1) reduce to their radial counterparts,

$$\hat{P} \rightarrow \hat{P}_{m+2} \equiv \partial_r^2 - \frac{1}{r}\partial_r + \frac{(m+2)^2}{r^2}, \quad (\text{C5})$$

$$\hat{Q} \rightarrow i\hat{Q}_{m+2} \equiv i \left[\frac{2(m+2)}{r}\partial_r - \frac{2(m+2)}{r^2} \right], \quad (\text{C6})$$

where the factor of i in the second line arises from $\partial_\theta \rightarrow i(m+2)$ acting on $e^{i(m+2)\theta}$. A direct evaluation of the angular projections in Eqs. (C2)-(C3) using Eq. (C4) then shows that $A_{m,m+2}^{jj'} = iB_{m,m+2}^{jj'}$, where B is real and given by

$$B_{m,m+2}^{jj'} = \frac{1}{2} \int_0^R r dr \Phi_{jm}(r) (\hat{P}_{m+2} + \hat{Q}_{m+2}) \Phi_{j',m+2}(r). \quad (\text{C7})$$

Thus, from here on we work with B alone. Substituting this relation into the off-diagonal entries of Eq. (A2) gives the off-diagonal block in the compact form

$$\mathbf{h}_{m,m+2}^{jj'} = -ta^2 B_{m,m+2}^{jj'} (\tau_x + i\tau_z), \quad (\text{C8})$$

where the sublattice matrix $\tau_x + i\tau_z$ mirrors the off-diagonal structure of the original Bloch Hamiltonian.

It remains to reduce the radial integral in Eq. (C7) to a closed form. We use the two standard Bessel recurrences,

$$J'_\nu(x) = \frac{\nu}{x} J_\nu(x) - J_{\nu+1}(x), \quad (\text{C9})$$

$$\frac{2\nu}{x} J_\nu(x) = J_{\nu-1}(x) + J_{\nu+1}(x), \quad (\text{C10})$$

together with the Laplacian eigenrelation (A6), where $\nabla_{m+2}^2 = \partial_r^2 + \frac{1}{r}\partial_r - \frac{(m+2)^2}{r^2}$ is the radial Laplacian in the $(m+2)$ channel. Adding and subtracting $\frac{1}{r}\partial_r$ and $\frac{(m+2)^2}{r^2}$ inside the differential bracket isolates the radial Laplacian, giving

$$B_{m,m+2}^{jj'} = \frac{1}{2} \int_0^R r dr \Phi_{jm}(r) \left[\nabla_{m+2}^2 + \frac{2(m+1)}{r}\partial_r + \frac{2(m+1)(m+2)}{r^2} \right] \Phi_{j',m+2}(r). \quad (\text{C11})$$

The eigenrelation replaces ∇_{m+2}^2 by $-(\beta_{j',m+2}/R)^2$, while the derivative recurrence (C9) at $\nu = m+2$ gives

$$\partial_r \Phi_{j',m+2}(r) = \frac{m+2}{r} \Phi_{j',m+2}(r) - \frac{\beta_{j',m+2}}{R} N_{j',m+2} J_{m+3}(\beta_{j',m+2} \frac{r}{R}), \quad (\text{C12})$$

with $N_{j',m+2} = \sqrt{2}/[RJ_{m+3}(\beta_{j',m+2})]$. One can verify that the J_{m+3} term generated here cancels exactly against the one produced when the $1/r^2$ term is re-expressed through the index recurrence (C10), so that only a J_{m+1} contribution survives. Collecting what remains,

$$B_{m,m+2}^{jj'} = -\frac{\beta_{j',m+2}^2}{2R^2} \int_0^R r dr \Phi_{jm}(r) \Phi_{j',m+2}(r) + \frac{\sqrt{2}(m+1)\beta_{j',m+2}}{R^2 J_{m+3}(\beta_{j',m+2})} \int_0^R dr \Phi_{jm}(r) J_{m+1}(\beta_{j',m+2} \frac{r}{R}). \quad (\text{C13})$$

Inserting the explicit forms of Φ_{jm} , rescaling to $x = r/R$, and applying the index recurrence (C10) once more to J_{m+2} cancels the intermediate J_{m+1} pieces between the two integrals and collapses them to a single Bessel integral of equal order,

$$B_{m,m+2}^{jj'} = \frac{\beta_{j',m+2}^2}{R^2 J_{m+1}(\beta_{jm}) J_{m+3}(\beta_{j',m+2})} \times \int_0^1 x J_m(\beta_{jm} x) J_m(\beta_{j',m+2} x) dx. \quad (\text{C14})$$

The remaining integral is a Lommel integral: for arguments $a \equiv \beta_{jm}$ not equal to $b \equiv \beta_{j',m+2}$ it reads

$$\int_0^1 x J_m(ax) J_m(bx) dx = \frac{b J_m(a) J_{m+1}(b) - a J_{m+1}(a) J_m(b)}{b^2 - a^2}. \quad (\text{C15})$$

where the first term in the numerator vanishes because $a = \beta_{jm}$ is a zero of J_m . For completeness, we also note the self-integral $\int_0^1 x J_m^2(ax) dx = \frac{1}{2} J_{m+1}^2(a)$, which is needed only in the exceptional case discussed below. Hence

$$B_{m,m+2}^{jj'} = \frac{b^2}{R^2 J_{m+1}(a) J_{m+3}(b)} \begin{cases} \frac{a J_{m+1}(a) J_m(b)}{a^2 - b^2}, & a \neq b, \\ \frac{J_{m+1}^2(a)}{2}, & a = b. \end{cases} \quad (\text{C16})$$

Applying $J_{m+3}(b) = -J_{m+1}(b)$, which follow from Eq. (C10) evaluated at the respective zeros, the $a \neq b$ case simplifies to

$$B_{m,m+2}^{jj'} = \frac{\beta_{jm} \beta_{j',m+2}^2 J_m(\beta_{j',m+2})}{R^2 J_{m+1}(\beta_{j',m+2}) (\beta_{j',m+2}^2 - \beta_{jm}^2)}. \quad (\text{C17})$$

For $m \neq -1$, the zeros of J_m and J_{m+2} are distinct, so Eq. (C17) is valid for all j and j' . The only exceptional case is $m = -1$, for which $J_{-1}(x) = -J_1(x)$ and $\beta_{j,-1} = \beta_{j,1}$. For $j = j'$, one has $a = b$, and the matrix element must therefore be evaluated from the second line of

Eq. (C16), yielding $B_{-1,1}^{jj} = -\frac{\beta_{j1}^2}{2R^2}$. The reverse coupling $\mathbf{h}_{m+2,m}$ follows from the same reduction with the two channels interchanged, $(m, j) \leftrightarrow (m+2, j')$. Hermiticity of $\mathbf{H}$ then follows directly, with $h_{m+2,m}^{j'j} = (h_{m,m+2}^{jj'})^*$ verified analytically from Eq. (C17) under the same interchange.

Appendix D: Pairing terms

For a singly quantized vortex with on-site, sublattice-uniform pairing $\Delta(\mathbf{r}) = \Delta(r)e^{-i\theta}$, the matrix elements between Bessel basis states follow directly from the angular selection rule of App. B,

$$\langle jm|\Delta(\mathbf{r})|j', m'\rangle = \delta_{m',m+1} C_{m,m+1}^{jj'} \tau_0, \quad (\text{D1})$$

where τ_0 reflects the on-site, sublattice-diagonal form of the pairing field, and the radial overlap is

$$C_{m,m+1}^{jj'} \equiv \int_0^R r dr \Phi_{jm}(r) \Delta(r) \Phi_{j',m+1}(r). \quad (\text{D2})$$

Hence $\Delta_{mm'} = \mathbf{0}$ unless $m' = m+1$, as already anticipated in Eq. (B8). The conjugate sector $\langle jm|\Delta^*(\mathbf{r})|j'm'\rangle$ similarly enforces $m' = m-1$ and produces the overlap $(C_{m-1,m}^{jj'})^*$ entering the hole equation. Together, the $m' = m+1$ block and its conjugate $m' = m-1$ counterpart constitute the full pairing matrix Δ and its Hermitian conjugate $\Delta^\dagger$ appearing in Eq. (B4).

Appendix E: Self-consistency relations

The self-consistency loop of Algorithm 1 is completed by expressing the gap and density equations (23)-(25) in

terms of the Bessel basis coefficients $\{u_{njm}^S, v_{njm}^S\}$. The key simplification is that every observable is a thermal average over the angular coordinate, $A(r) = \frac{1}{2\pi} \int_0^{2\pi} d\theta A(\mathbf{r})$, so each angular integral reduces to a Kronecker delta via Eq. (A4), fixing the angular-momentum bookkeeping in each case.

For the gap equation, substituting the expansion (B1) into Eq. (23) and stripping the vortex phase $\Delta(\mathbf{r}) = e^{-i\theta} \Delta(r)$ leaves an angular factor $e^{i(-m-1+m')\theta}$ under the average, which Eq. (A4) collapses to $m' = m+1$. The radial gap profile then reads

$$\Delta(r) = -\frac{U}{2\pi} \sum_{njj'm} u_{njm}^S v_{nj',m+1}^{S*} \Phi_{jm}(r) \Phi_{j',m+1}(r) f(E_n). \quad (\text{E1})$$

The structure of Eq. (E1) is consistent with the pairing block of the BdG matrix (B8): particle components at angular momentum m couple to hole components at $m+1$. For the density equations, the spin-resolved densities are scalars, so the angular average simply sets $m = m'$. Using $f(-E_n) = 1 - f(E_n)$ and inserting Eq. (B1) into Eq. (25) gives

$$\rho_\uparrow(r) = \frac{1}{2\pi} \sum_{njj'mS} u_{nj'm}^{S*} u_{njm}^S \Phi_{jm}(r) \Phi_{j'm}(r) f(E_n), \quad (\text{E2})$$

$$\rho_\downarrow(r) = \frac{1}{2\pi} \sum_{njj'mS} v_{nj'm}^{S*} v_{njm}^S \Phi_{jm}(r) \Phi_{j'm}(r) f(-E_n), \quad (\text{E3})$$

where $f(-E_n) = 1 - f(E_n)$. The total local density feeding the algorithm is $\rho(r) = \rho_\uparrow(r) + \rho_\downarrow(r)$, summed over the sublattice index. Equations (E1)-(E3) are the explicit basis representations evaluated at each iteration of the self-consistent BdG solver, with the updated $\Delta(r)$ feeding back into the radial overlaps $C_{m,m+1}^{jj'}$ of Eq. (D2).

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